File systems:-

File: collection of logically related entities (records)

File is having various Attributes

(i) Name
(ii) Type
(iii) Location
(iv) Size
(v) Creation date
(vi) Last modified date
(vii) permissions
(viii) Owner
(ix) password.

- All the attributes of the file is called as "File context"
- File context will be stored in the F.C.B (File control Block)
- File is having various types:-
  - . doc
  - . Txt
  - . bat
  - . obj
  - . exe
  - . dll
  - . pdf
  - . Jpg
  - . png
  - . c /. Cpp /. Java /. Html / . Php
  - . mp3 | . mp4
  - . mkv
  - . avi
  - . xls | . xlsx
  - . apk

various operations performed on the file

i. create
ii. open
iii. Write
iv. append
v. truncate
vi. save
vii. save as
viii. Rename

ix. Send
x. Copy
xi. Paste
xii. cut
xiii. print
xiv. delete
xv. close

Interview Question:

C:\ABC\XYZ| Kolkata.doc

C:\PQR\MNO| Kolkata.doc

(i) Copy + Paste → It reads the entire file from the source and writes into the destination
(ii) Cut + paste.

- Copy + paste will take more time
- Cut + paste will take less time because it will link the address
- Only valid on the same drive.
- It only updates the file system pointers, not the actual content.
- For cross-drives moves both will take same time

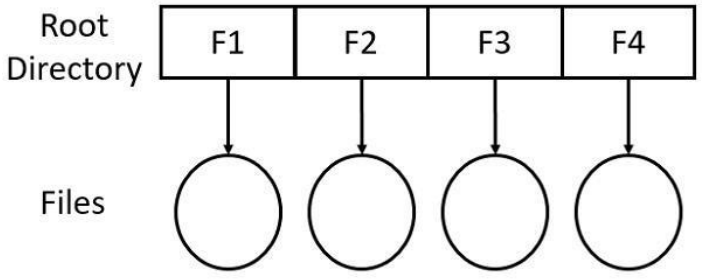
File has various access methods.

i. Sequential access
ii. Random access.

For the better classification of the files, files will be stored in the directory.

Directory structure :-

(i) single level directory:-



a. Implementation of directory structure is easy
b. Two files can not have the same name .
c. Searching time for the specifical file will be more.

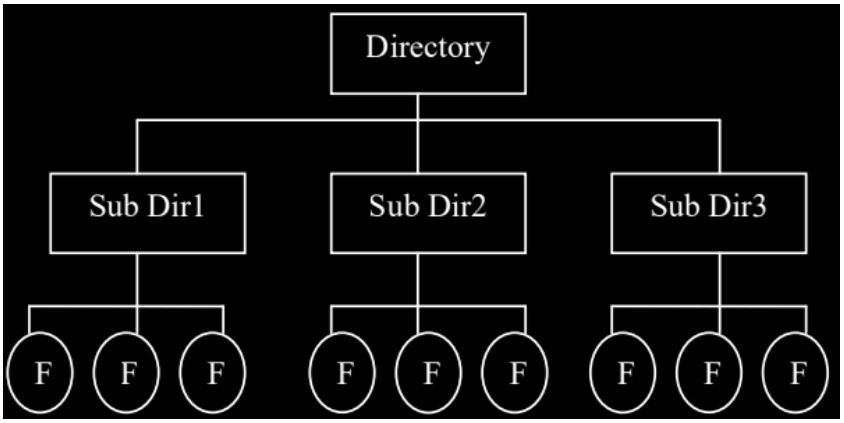
(ii) Multi level directory

or

Tree level directory

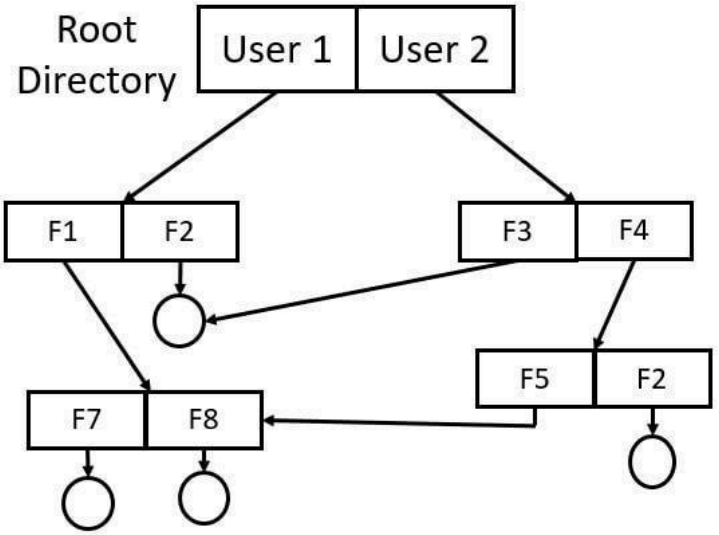
or

Hierarchical level directory :-



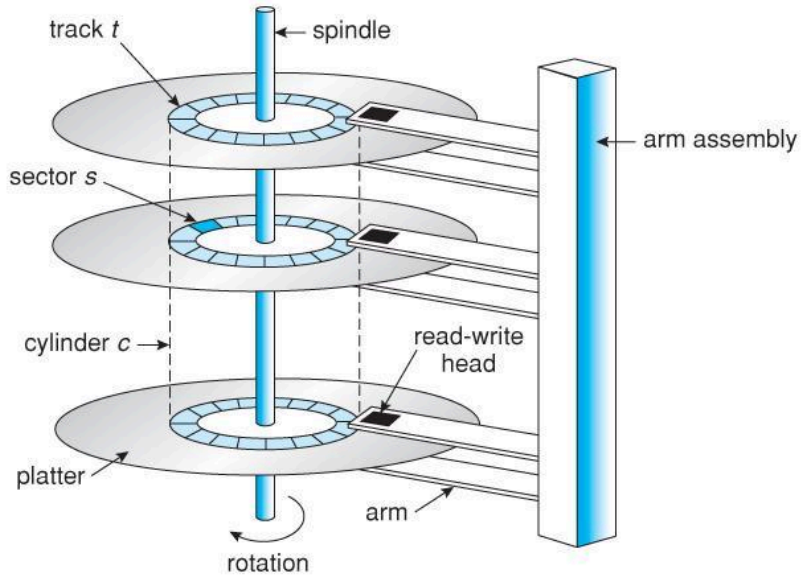
- Implementation of directory structure is difficult,
- Better classification of file as per criteria
- Searching time for the specific file will be Less.
- If the same file exists in the two different Sub directories then if one file is updated then the other file has to be updated accordingly Otherwise there will be an inconsistency.

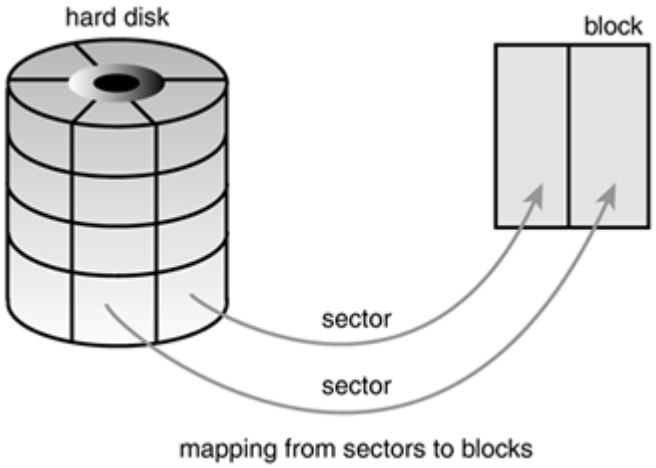
(iii) A cyclic graph directory:



a. Implementation of directory structure is difficult or complicated
b. Better classification of the files as per the criteria.
c. Searching time for the specific files will be less.
d. If the same file exists in the two different directories then the other file will be updated automatically -----file sharing concept.

Disk structure





mapping from sectors to blocks

Question: Consider a disk which has 16 platters and each platter has two surfaces and each surface is divided into 1K tracks and each track is further divided into 512 sectors and each sector can store 2 KB data, then calculate.

a. capacity of the disk?
b. How many many bits are required to identify a specific sector of the disk?

Solution :

(a)

$$2^4 \times 2^1 \times 2^{10} \times 2^9 \times 2^{11} B$$

$$\Rightarrow 2^{35} B = 2^5 \times 2^{30} B$$

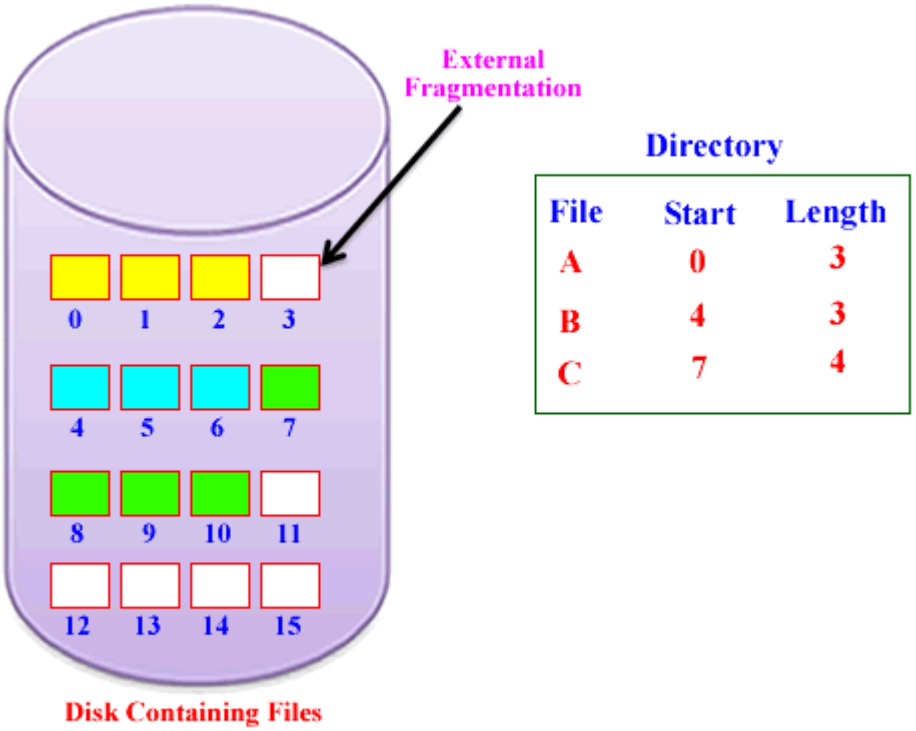
$$= 32GB.$$

(b)

$4 + 1 + 10 + 9 = 24$ bits are required to identify the specific sector of bits.

Disk space allocation methods. –

(1) Contiguous allocation:-

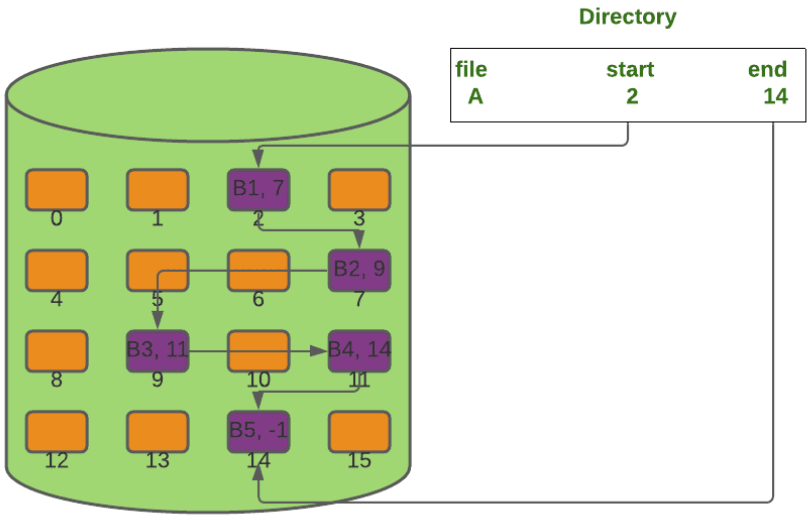


a. In the contiguous allocation file allocation whenever the file is created, the disk blocks are allocated in Continuous manner.
b. Every file is associated with 2 parameters
    - Starting DBA
    - size
c. Increasing the file size may not always be possible.
d. It is suffering from external fragmentation.
e. Internal fragmentation may exist in the last disk block of the files
f. It supports both sequential and random access of the file

    Starting D.B.A + offset = Random Access

    Eg: $9 + 4 \Rightarrow 13$

(2) Linked allocation (or) Non contiguous allocation:-

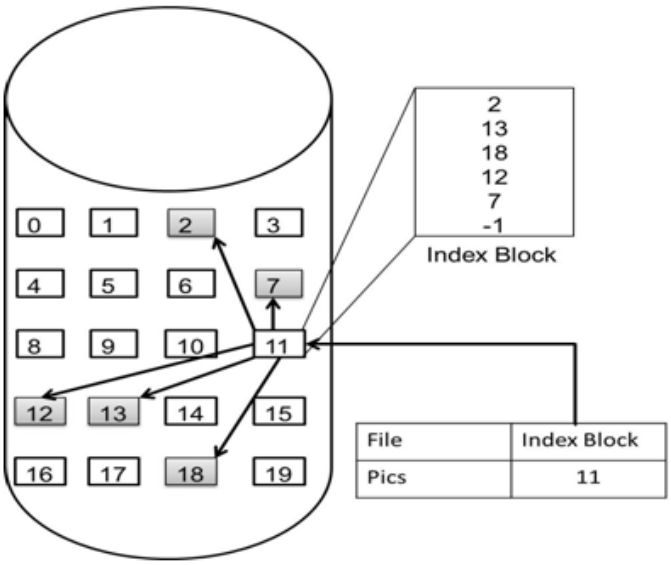


File	Starting DBA	Ending DBA
A	2	14

In the linked allocation whenever the file is created, disk blocks are allocated in a non- Contiguous manner.

- Every file is associated with the two parameters
  Starting D.B.A.
  Ending D.B.A
- Increase in the file size is always possible if the free disk block is available
- There is no external Fragmentation.
- Internal fragmentation may exist in the last disk block of the file.
- There is an overhead of maintaining the pointer in every disk block.
- If the pointer of any disk block is lost the file will be truncated.
- It supports only sequential access to the file.

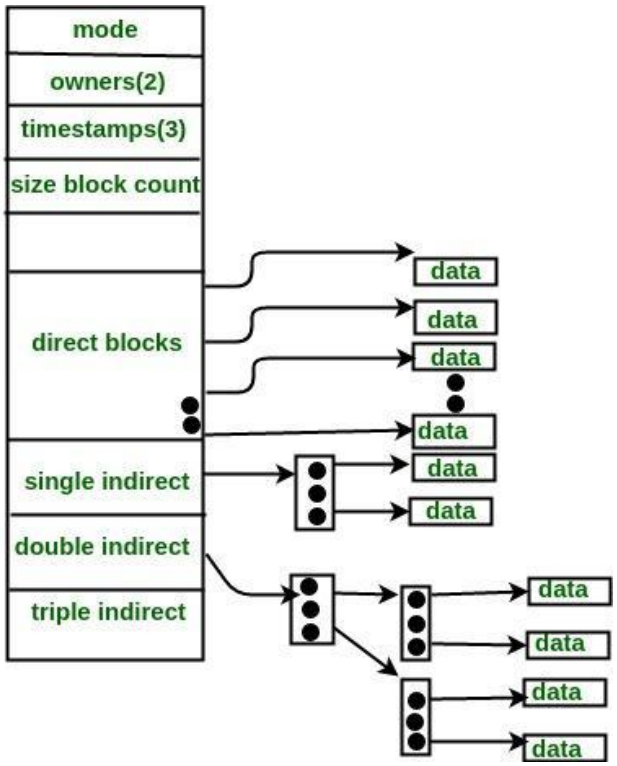
(3) Indexed Allocation



File	Indexed Node
pics	11

- In the indexed allocation, every file is associated with its own indexed Node.
- Indexed node contains all the disk block addresses of the file.
- Disk blocks are allocated in the non- contiguous manner.
- There are no external Fragmentations.
- Internal fragmentation exists in the last disk block.
- If the file size is very very large than one disk block may not be sufficient to store all the disk block addresses of the file.
- If the file size is very very small then it is a waste of using the entire one disk block Just to store the addresses.

UNIX - I. Node Implementation
(An Extension of indexed allocation)



- Unix follows I-Node way of implementation In order to allocate disk space to the files
- Every file is associated with the its own I-Node
- Whenever the file is created depending on the size of the file, the file will be stored only in one particular place, may be- In the direct disk block address or may be in the single indirect and so on.
- $\frac{D.B\ size}{D.B.A}$ = # of D.B.A's one D.B. can contain

Question. consider the Unix I Node which has 10 direct D.B.A , 1 single indirect, 1 double indirect and 1 triple indirect disk block addresses . DB size = 1KB, and DBA size = 4B. Then what is the max file size possible ?

Answer:-

Maximum File Size in Unix I-Node Structure

Given:

- Disk Block size (DB size) = 1 KB

- Disk Block Address size (DBA size) = 4 Bytes

Calculations

Number of DBAs per block:

1024 Bytes / 4 Bytes = 256 DBAs per block

1. Direct Blocks

10 blocks × 1 KB = 10 KB

2. Single Indirect Block

256 blocks × 1 KB = 256 KB

3. Double Indirect Block

256 × 256 × 1 KB = 65,536 KB = 64 MB

4. Triple Indirect Block

256 × 256 × 256 × 1 KB = 16,777,216 KB = 16,384 MB = 16 GB

Final Maximum File Size

10 KB + 256 KB + 64 MB + 16 GB ≈ 16 GB